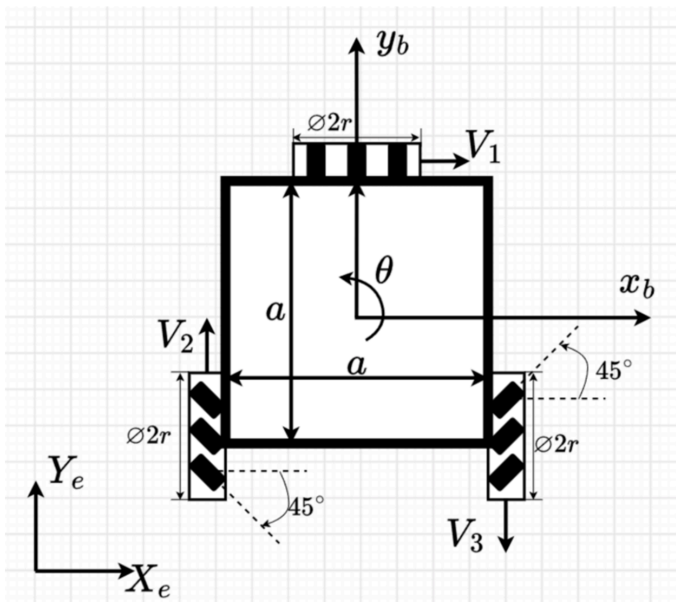
Inverse Kinematics of a Three-Wheel Omnidirectional Robot

ME597 - Autonomous Systems - Fall 2026

A planar mobile robot has two mecanum wheels with roller angles of magnitude 45° and one omni wheel. The body-fixed frame $\{B\} = (x_b, y_b)$ is at center of the robot, and the inertial frame is $\{I\} = (X_e, Y_e)$. The robot has a dimension of length a and width a. All wheels have radius r. The wheel angular velocities ω_1, ω_2, and ω_3 are defined as positive when no-slip wheel rotation drives the corresponding wheel center in the directions indicated by V_1, V_2, and V_3, respectively. At the instant of interest, the body-fixed frame is aligned with the inertial frame, so $\theta = 0$.



The passive rollers permit free motion parallel to their axes and impose a no-slip rolling constraint perpendicular to their axes. The following equation can be derived from this condition for wheel i, where l_i is the distance from the body-frame center to the center of wheel i, α_i is the signed angle from x_b to the wheel-position vector, β_i is the signed angle from the wheel-position vector to the main wheel axle, and γ_i is the signed angle from the wheel plane to the roller axle:

$$[\ \sin(\alpha_i + \beta_i + \gamma_i) \quad -\cos(\alpha_i + \beta_i + \gamma_i) \quad -l_i \cos(\beta_i + \gamma_i)\ ] R(\theta) \dot{\xi}_I = r\omega_i \cos\gamma_i,$$

where

$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

maps inertial-frame velocity components into the body frame.

Using the rolling constraint, express the robot's inertial frame velocity $(\dot{X}_e, \dot{Y}_e, \dot{\theta})^\top$ as a function of the rotation velocity of the wheels $(\omega_1, \omega_2, \omega_3)^\top$.